

AERO-GUARD Prototype – Python Tech Stack

Backend

Python 3.12+, FastAPI, Uvicorn

Frontend

Streamlit (recommended for prototype). Alternative: React + Tailwind CSS.

Database

SQLite for prototype, PostgreSQL for production.

Authentication

JWT Tokens with role-based access (Provider, Manager, Helpdesk, Consultant).

OCR

EasyOCR or Pytesseract for passport data extraction.

Analytics

Plotly for ADM trends, violations, and revenue protection dashboards.

Browser Extension

Simple HTML/CSS/JavaScript extension that opens the AERO-GUARD dashboard.

Provider Features

Agency provisioning, violation monitoring, ADM reports, voucher management, helpdesk support.

Consultant Features

Real-time alerts, ADM dashboard, passport upload, helpdesk chat, voucher notifications.

Database Tables

Users, Agencies, Violations, ADM Audits, Vouchers.

Folder Structure

aero_guard/ backend/ frontend/ ocr/ reports/ uploads/ requirements.txt

Recommended MVP

Login, Provider Dashboard, Violation Dashboard, Consultant Dashboard, Passport Upload.